

Air Blower engineering manual

Estimate motor input power and compression efficiency; track bearing condition and flow loss against a reference period.

Run the Python model

Open `air_blower_engine.py` in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains `schemaVersion`, `model`, `source`, `timezone`, `config` and `measurements`. Each measurement contains `timestamp`, `values` and optional `quality`. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs · 6 rows

INPUT	UNIT	USE
Timestamp; motor current for each train	date/time; A	Select the running train and calculate power.
Suction / discharge pressure for each train	kPaa / kPag	Calculate absolute pressure ratio and pressure rise.
Total air flow; suction and discharge gas temperature	Nm³/h; °C	Flow comparison and thermodynamic efficiency. Both temperatures are needed.
Vibration; bearing temperature for each train	mm/s RMS; °C	Condition alerts and recent trends.
Bypass opening; filter differential pressure; pressure setpoint	% open; bar; kPag	Operating conditions and baseline eligibility.
Voltage, power-factor method, k, atmosphere, limits	Configuration	Use values for the installed machine.

Calculations

PRESSURE

$$P_1 = \text{suction} / 100; P_2 = \text{discharge} / 100 + P_{\text{atm}}; r = P_2 / P_1$$

Pressures in bar absolute; $\Delta P = P_2 - P_1$.

MOTOR INPUT

$$P_e = \sqrt{3} \times V \times I \times PF / 1000$$

kW. PF is interpolated from the motor table or fixed. This is electrical input, not shaft power. Total power sums both running motors; shared flow is not assigned to one train when both run.

ISENTROPIC EFFICIENCY

$$\eta_s = 100 \times T_1 \times [r^{((k-1)/k)} - 1] / (T_2 - T_1)$$

Temperatures in kelvin. Ideal-gas estimate using compatible inlet/outlet measurements.

POLYTROPIC EFFICIENCY

$$\eta_p = 100 \times [(k-1)/k] \times \ln(r) / \ln(T_2/T_1)$$

Requires positive absolute pressures and a temperature rise. Results above 100% require data review.

FLOW LOSS

$$Q_{\text{expected}} = a + bI; \text{loss} = \max(0, 100 \times (Q_{\text{expected}} - Q) / Q_{\text{expected}})$$

a and b are least-squares fits for each train. The initial configured period is assumed healthy. Comparisons begin after that period and require complete inputs within its observed current and pressure-ratio range; constant speed is assumed.

BEARING AND VIBRATION TREND

$$\text{slope} = \Sigma[(t - \bar{t})(x - \bar{x})] / \Sigma(t - \bar{t})^2$$

Up to seven finite, fresh observations from the same train within seven days. Bearing days to advisory = (limit - temperature) / positive slope.

OPERATING DEVIATION

$$D = 100 \times \sqrt{[(\delta Q)^2 + \delta r^2 + (\text{bypass}/100)^2] / 3}$$

Fractional deviations from design flow and pressure ratio. Informational only; no axial-force or failure prediction.

NORMAL-FLOW REFERENCE

$$Q_n = (\text{mass flow} / \text{molecular weight}) \times 22.414$$

kg/h and kg/kmol give Nm³/h at 0 °C and 101.325 kPa. 26,665 / 29 gives about 20,609 Nm³/h; match the meter's wet/dry basis.

FLOW-PRESSURE INDICATOR

$$\text{Index power} = Q_n \times \Delta P / 36; \text{index} = 100 \times \text{index power} / P_e$$

Legacy screening index using normal flow. It is not compressor gas power or thermodynamic efficiency.

Useful trends

- Measured and expected flow, with flow-loss percentage.
- Motor input power, pressure rise and thermodynamic efficiency at comparable loads.
- Bearing temperature and vibration against configured limits; filter pressure drop and bypass opening.

Assumptions and limits

- Automatic selection evaluates one train; if both run, it selects the higher-current train. Total electrical input sums running motors. Shared flow is not allocated to an individual train when both run or the other current is missing.
- The reference period is assumed healthy. Bypass and filter checks alone do not prove comparable speed, gas conditions or load; no current or pressure-ratio extrapolation is used, and speed is assumed constant.
- Motor PF interpolation assumes the supplied motor at steady operation and rated voltage. Outside the table, endpoint PF is held; startup power is not represented.
- Bearing alerts and trend projections do not predict remaining life. Direct axial-position or thrust-bearing measurements are needed for thrust assessment.
- Thresholds are configurable demonstration values, not certified ISO zones or shutdown settings. A bounded held suction temperature is labelled; simulated signals do not establish actual efficiency or health.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- NASA Glenn: compressor thermodynamics
- NIST: linear least squares
- ISO 20816-3: vibration evaluation scope
- Supplied blower and motor datasheets: design points and power-factor table